

A telescoping obstruction to coordinate cycles in beam observability

A candidate full answer to Open Question (iv) of arXiv:1903.07804

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Status: candidate full result, likely valid. The two-segment inverse observability inequality holds for every pair of distinct non-integer rational velocities. More strongly, it holds for every pair of distinct non-integer real velocities.

1 Source problem

Jaming and Komornik study moving observations of the periodic beam equation [1]. Their Open Question (iv), on PDF page 19, asks whether their inverse inequality (4.8) holds for any two distinct non-integer rationals a_1, a_2 .

$$\{(m, m^2 + n^2) : m, n \in \mathbb{Z}\},$$

and this was proved in [22].

□

6. OPEN PROBLEMS

We end this paper by a list of open questions:

- (i) We do not know whether the inverse inequality in Theorem 3.2 (ii) fails for every choice of finitely many segments (x_i, t_i) instead of only one segment.
- (ii) Can we consider more general segments in Proposition 3.3 (ii)?
- (iii) How to modify Proposition 3.3 for Neumann boundary conditions?
- (iv) We have no examples for a cycle in Theorem 4.8. Does the inverse inequality (4.8) of Theorem 4.8 hold for any two different non-integer rationals a_1, a_2 ?
- (v) If there are examples of cycles in the preceding question, then does there exist an integer $m \geq 3$ such that the inverse inequalities

$$\sum_{k \in \mathbb{Z}} (|c_k^+|^2 + |c_k^-|^2) \ll \sum_{i=1}^m \int_0^T |u(t_i + t, x_i - a_i t)|^2 dt$$

hold for all choices of different nonzero integers $a_1, \dots, a_m$?

REFERENCES

Figure 1: Open Question (iv) and its context in [1, p. 19].

For reference, a solution has the Fourier representation

$$u(t, x) = c_0^+ + c_0^- t + \sum_{k \in \mathbb{Z} \setminus \{0\}} \left(c_k^+ e^{i(k^2 t + kx)} + c_k^- e^{i(-k^2 t + kx)} \right).$$

Given $T > 0$, base points (t_j, x_j) , and velocities a_j , the requested inverse inequality is

$$\sum_{k \in \mathbb{Z}} (|c_k^+|^2 + |c_k^-|^2) \leq C \sum_{j=1}^2 \int_0^T |u(t_j + t, x_j - a_j t)|^2 dt. \quad (1)$$

2 Proof intuition

The only obstruction left by the source's nonharmonic Fourier estimates is a finite cycle of coincident frequencies. Such coincidences are encoded by lattice points on the circles

$$S_a = \{(x, y) \in \mathbb{R}^2 : x^2 - ax + y^2 + ay = 0\}.$$

In a cycle between two non-integral parameters $a \neq b$, the lattice points can be written alternately as $(x_i, y_i) \in S_a$ and $(x_{i+1}, y_i) \in S_b$. Subtracting the paired circle equations and summing once around the cycle makes all quadratic terms telescope. This forces the sums of the two coordinates to agree. Summing just the S_a equations then forces a sum of squares to be zero, which is impossible because the origin is not a graph vertex.

3 The finite collision graph

Put

$$A_a = (S_a \cap \mathbb{Z}^2) \setminus \{(0, 0)\},$$

and let A_a^+ and A_a^- be its first- and second-coordinate projections. The set A_a is finite. Moreover, it is empty if a is irrational, because every $(x, y) \in A_a$ satisfies

$$a = \frac{x^2 + y^2}{x - y} \in \mathbb{Q}. \quad (2)$$

For distinct parameters a, b , form a graph whose vertices are $A_a \cup A_b$, joining two vertices when they share a coordinate. The source's estimate (4.9), with the harmless observation phases left explicit, has the form

$$\begin{aligned} \int_0^T |u(t_j + t, x_j - a_j t)|^2 dt &\asymp \sum_{k \notin A_{a_j}^+} |c_k^+|^2 + \sum_{m \notin A_{a_j}^-} |c_m^-|^2 \\ &\quad + \sum_{(k, m) \in A_{a_j}} |\alpha_{j, k} c_k^+ + \beta_{j, m} c_m^-|^2, \end{aligned} \quad (3)$$

where all $\alpha_{j, k}, \beta_{j, m}$ are unimodular. Thus the only finite-dimensional obstruction to (1) is a closed chain of collision relations.

Lemma 1 (coordinate injectivity). *If $a \notin \mathbb{Z}$, no two distinct points of A_a share a first coordinate or share a second coordinate.*

Proof. If $(x, y), (x', y) \in A_a$ with $x \neq x'$, then x, x' are the two roots of

$$X^2 - aX + y^2 + ay = 0.$$

Hence $a = x + x' \in \mathbb{Z}$, a contradiction. If $(x, y), (x, y') \in A_a$ with $y \neq y'$, the corresponding quadratic in Y gives $-a = y + y' \in \mathbb{Z}$. $\square$

Proposition 1 (the graph reduction). *Let a, b be distinct non-integers. If the coordinate-sharing graph on $A_a \cup A_b$ has no cycle, then (1) holds with $(a_1, a_2) = (a, b)$ for arbitrary base points and every $T > 0$.*

Proof. If either lattice set is empty, the one-observation result [1, Theorem 4.4(ii)] already proves the claim. Otherwise, Lemma 1 and (2) show that the graph is bipartite, has degree at most two, and hence is a disjoint union of paths when it has no cycle.

Consider the sum of the two quadratic forms in (3). If it vanishes, an unshared coordinate at an endpoint of any path is zero because it occurs in one of the first two sums. The endpoint collision relation then forces the next coordinate variable to vanish; induction propagates zero along the whole path. Isolated vertices are handled in the same way. All variables in the finite collision block therefore vanish, while the first two sums already control every variable outside that block. The resulting finite-dimensional quadratic form is positive definite, hence coercive. Together with (3), this is (1). $\square$

4 No coordinate cycle can occur

Lemma 2 (telescoping cycle obstruction). *For distinct $a, b \in \mathbb{R} \setminus \mathbb{Z}$, the coordinate-sharing graph on $A_a \cup A_b$ has no cycle.*

Proof. Suppose a cycle exists. A nonzero lattice point determines its parameter uniquely by (2), so $A_a \cap A_b = \emptyset$. Lemma 1 forces both the parameter labels and the horizontal/vertical directions to alternate around the cycle. After choosing an orientation, its vertices may therefore be labeled cyclically as

$$P_i = (x_i, y_i) \in A_a, \quad Q_i = (x_{i+1}, y_i) \in A_b, \quad i = 1, \dots, r,$$

where $x_{r+1} = x_1$. Their circle equations are

$$x_i^2 - ax_i + y_i^2 + ay_i = 0, \tag{4}$$

$$x_{i+1}^2 - bx_{i+1} + y_i^2 + by_i = 0. \tag{5}$$

Subtract (4) from (5) and sum over i . The square terms telescope cyclically, leaving

$$(a - b) \left(\sum_{i=1}^r x_i - \sum_{i=1}^r y_i \right) = 0.$$

Since $a \neq b$,

$$\sum_{i=1}^r x_i = \sum_{i=1}^r y_i. \tag{6}$$

Finally, sum (4) over i and use (6):

$$\sum_{i=1}^r (x_i^2 + y_i^2) = a \left(\sum_{i=1}^r x_i - \sum_{i=1}^r y_i \right) = 0.$$

Every x_i and y_i is zero. This says every P_i is the excluded origin, a contradiction. $\square$

Theorem 1. *For every two distinct $a_1, a_2 \in \mathbb{R} \setminus \mathbb{Z}$, every $T > 0$, and arbitrary base points $(t_1, x_1), (t_2, x_2) \in \mathbb{R}^2$, the periodic beam solutions satisfy the inverse observability inequality (1). In particular, Open Question (iv) of [1] has an affirmative answer.*

Proof. Combine Proposition 1 with Lemma 2. $\square$

5 Checks, scope, and novelty

The proof uses rationality nowhere. If a parameter is irrational, its lattice set is empty; otherwise (2) makes it rational automatically. Thus the theorem strictly strengthens the source’s rational question. It does not address integer velocities, for which coordinate injectivity fails.

The arXiv source prints “integers” in Lemma 4.7 and Theorem 4.8. This is inconsistent with the lemma’s proof, which derives integrality as a contradiction, with the subsequent non-integral rational example, and with Open Question (iv). Proposition 1 states the intended non-integer reduction directly from the source’s estimate (4.9), so the result does not rely on that printed word.

As a non-proof regression check, the included script exhaustively found no cycle for all parameters of absolute value at most 120 (21,178 parameter classes and 3,044 candidate pairs). A bounded search on 13 August 2026 used the exact question, title, arXiv id, graph notation, and core cycle and observability phrases. The current arXiv version still states the question as open, and no later explicit answer or matching telescoping argument was found. Novelty confidence is moderate.

Human-review recommendation. Check the alternating labeling and the two telescoping sums, the endpoint propagation in the phase-weighted finite forest, and the editorial integer/non-integer issue in the source.

References

- [1] P. Jaming and V. Komornik, *Moving and oblique observations of beams and plates*, Evolution Equations and Control Theory **9** (2020), no. 2, 447–468; arXiv:1903.07804.